# Course Project

## INT2067 Introduction to Programming and Problem Solving

## 2026/27 Semester 1

## 1. Submission (Actual Date will be specified in Moodle's submission link)

Each group is required to submit the following by the stated deadlines. Only one member in a group needs to submit the group works. **Late submission may not be accepted.**

**[Week 5] Project Proposal.** Briefly (about 200 words) describe what you will attempt to do in pdf file format Submit to Moodle Detail write down the group No, all members name and students ID

**[Week 10] Source Code and Test Cases and Presentation PPT.**

- Name your Python file as project.py.
- Your test cases should be stated in README.md.
- Presentation PowerPoint to be presentation.pptx
- Zip all files (Python files, data files, and readme file) into one file named **groupN_project.zip.**
- Submit your project code and test cases on Moodle.

**[Week 12] Individual Report.** Submit your individual report named **report.zip** (in pdf format) on Moodle.

## 2. Group Formation

Each group should consist of not more than **5 students**.

## 3. Project Specification

This project requires your group to develop a **Python program** that solves a **simple "real-world application"** relevant to your interests. It should focus on a **single purpose**—for example: a personal contact manager, a text-based computer game, a student assessment advice system, self-learning management system, a financial advises system etc… The **major evaluation criteria** for this project will include:

- The difficulty of the chosen topic.
- The design work
- Programming quality and skills demonstrated
- Testing cases elicitation and strategies
- The overall development process

Students are strictly required to apply the concepts learned in this course. Your project needs to be developed mainly include the following:

1. Basic program components
2. All Java control features, e.g., iterations, selective, and decision controls
3. The use of methods/functions to organize the codes in the project
4. Suitable Python data structure, like List, Dictionary, String, and Class/Objects
5. File operation

**NOTE: The application must be text-based, without any graphical user interface (GUI).**

## 4. Program Execution and Test Cases

Your project needs to be tested by running Python in a terminal. Your source code should be named project.py. If your project applies files operations, please make sure you use suitable (e.g., relative paths), and the files are put in correct location. If use Google CoLab to develop the project, make sure the file are suitable amounted while running in VS Code, or other CoLab platform. For use Thonny, please grantee the files are at the correct location to your program.

You are advice to design a set of test cases that are able to illustrate the important features of your project application, as well as shown in the presentation. During running the program in presentation, please show up the steps as a sample session. For example from login to a completed operation.

## 5. Plagiarism Policy

Plagiarism in any form is strictly prohibited and will result in penalties. All work submitted must be original and created by the group members. If you use external resources (e.g., 3rd-party libraries, or open-source code), you must properly cite them in your report and ensure they comply with the project requirements. Each student must write their own Individual Report independently. Reports should not be copied from other group members or external sources.

Besides, you need to write-down a declaration on top of the program as follows:

‘’’

**This code is my own individual and original work with very limited AI assistant. For any part of codes that are referenced to, supported by AI, acknowledgement will be specified**

‘’’

Inside the program, the acknowledgement will look like as follows shown in the corresponding codes:

**# Acknowledgement: I have asked Thomas, Gordon, and Emily for help.**

**# Acknowledgement: I have used AI tools (such as ChatGPT, gpt4o via POE, etc.) in working on this assignment.**

**Warning:** Any violation of this policy may lead to deduction of marks or a failing grade for the project, or disciplinary action according to the institution's academic integrity policies.

# 6. Evaluation Criteria

## 1. Originality and practicality (10%)

The idea of your project is **original and innovative**, including some **'smart features'.** Your application should be practical and applicable to real-world scenarios. Additionally, it would be beneficial to explain how useful the application could be in your presentation and report.

## 2. Difficulty (25%)

Your project will be evaluated based on the level of complexity by course-related techniques incorporated into your application. The table below outlines the expected difficulty levels and corresponding Python techniques:

| Grade | Expected Techniques in Python Application |
|---|---|
| Minimum | Basic output with meaningful content. |
| Marginal | Input, output, and basic computation. |
| Satisfactory | Input, output, computation, and conditionals or iterations. |
| Good | Input, output, computation, conditionals or iterations, and data structures, file IO, and/or classes. |
| Excellent | Input, output, computation, conditionals, iterations, data structures, file IO, and classes, and apply modulization concept |

## 3. Correctness and Usability (25%)

Your application should satisfy the following requirements:

- **Proper Execution** – The program must be fully runnable without crashes. It should be executed correctly in the standard Python environments.
- **Ease of Use** – The application should have a clear and intuitive command-line interface, allowing users to interact with it without confusion. Instructions should be provided where necessary.
- **Good Test Cases** – The application should provide test cases to illustrate the important features of your program in a README.md file.
- **Error Handling** – The program should handle invalid inputs gracefully, providing meaningful error messages instead of crashing.
- **Responsiveness** – The application should provide timely feedback to the user, ensuring there are no unnecessary delays in processing input and displaying output.

## 4. Group Presentation (not over 15 minutes) (25%)

The presentation should have a logical structure, including an introduction, problem description, methodology, demonstration, and conclusion. The team should discuss the key technical aspects of the project, such as the programming concepts and techniques used.

You may use this presentation to show 'HOW GOOD of you Project' by focusing on the following.

1. Highlight the **difficulty level of your project** (e.g., have use class, or other complicated features upon the use of Python techniques)
2. Selecting suitable test cases to show-up your program is the **criteria of correctness and usability**

Every group member is required to be involved in the presentation. Absent and missing to presented member will give 0 mark in this part.

## 5. Individual Report (15%)

Each individual member is required to write a short report (Not more than 1,200 words) to describe and explain his/her own work. It needs to clearly describe your individual role and contributions to the project, specifying which tasks you worked on and how you contributed to the final product. The report should follow a logical format, including an introduction, project description, personal contributions, challenges faced, how it can be improved. Conclude with the lesson learned, and how your project's quality could be improved.